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The Foram Stress Index: A new tool for environmental assessment of soft-bottom environments using benthic foraminifera. A case study from the Saronikos Gulf, Greece, Eastern Mediterranean

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ABSTRACT

Saronikos Gulf, including the industrial zone of Elefsis Bay and the Port of Piraeus, is one of the most anthropogenically impacted coastal regions of Greece. Distinct assemblages of benthic foraminifers in sediment samples, collected from this gulf in February 2012, defined three zones that reflect abiotic parameters of the sediments (e.g., organic carbon, metal content). A low-diversity assemblage, dominated by stress-tolerant *Ammonia tepida* and *Bulimina* spp., was characteristic of samples from Elefsis Bay. Samples from the western and central part of Saronikos Gulf were the most variable with respect to both abiotic parameters and the foraminiferal assemblage, characterized by a mix of stress-tolerant and more sensitive taxa, especially *Bolivina* spp. and *Nonion fabum*. Samples from the coast of Salamis and at the eastern sector of the gulf were characterized by a diverse assemblage that included *Peneroplis pertusus*, miliolids, and a variety of small, epiphytic rotaliid taxa. A new biotic index, the Foram Stress Index (FSI), is based on the relative percentages of two ecological groups of benthic foraminiferal species, grouped according to their tolerance/sensitivity to organic matter enrichment and weighted proportionately to obtain a formula to define five ecological-status classes. The FSI produced three rankings for these samples (Poor, Moderate and Good), that strongly correlate with the macroinvertebrate-classification tool known as the BENTIX Index. The FSI provides a new tool to assess sediment or substrata quality based upon the benthic foraminiferal assemblages, which are a significant component of living meiobenthic communities that are generally not considered in most biotic benthic indices.

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1. Introduction

Benthic foraminifera are shelled, unicellular, eukaryotic microorganisms that are generally diverse and abundant in marine ecosystems. Foraminifera respond directly to stimuli from their environment (Goldstein, 1999) and their life spans of weeks to months make their populations and assemblages sensitive indicators of environmental fluctuations (Murray, 2006), when compared to benthic macrofauna. Hence, changes in benthic faunal abundance, species diversity and variation in morphology of tests provide evidence for changes in environmental parameters such as flux of organic matter, oxygen availability, bottom substrata and water depth. Moreover, the occurrence of selected benthic foraminiferal species can reflect the state of the ecosystem (e.g., Boltovskoy et al., 1991; Geslin et al., 1998; Yanko et al., 1999; Debenay et al., 2000). Therefore, benthic foraminifera are reliable bio-indicators of natural environmental and anthropogenic impact/pollution, and are increasingly used in the monitoring of environmental health and balance in coastal and marginal marine ecosystems (e.g., Alve, 1995; Yanko et al., 1999; Schafer, 2000; Samir and El-Din, 2001; Armynot du Châtelet et al., 2004; Ferraro et al., 2006; Mojtahid et al., 2006, 2008; Frontalini et al., 2009; Carnahan et al., 2009; Debenay and Fernandez, 2009; Denoyelle et al., 2010; Koukousioura et al., 2011; Donnici et al., 2012; Valipour et al., 2012, 2013a,b; Martins et al., 2013; Barras et al., 2014).

Most benthic biotic indices used in ecological quality assessment have been developed and experimentally designed to assess soft bottom macrofauna, excluding the benthic meiofauna compartment of the ecosystem. Meiofauna is less often examined in routine procedures, because analysis involves special laboratory

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procedures. Thus foraminifers, which are components of the meiofauna, can provide sensitive indicators of environmental stress that reflect the condition of this important component of an ecosystem.

Shallow marine ecosystems in urban coastal areas are heavily affected by human activities (domestic and industrial effluents, aquaculture, and tourism) that degrade their quality. One of the most human-impacted marine environments in Greece (e.g., Kapsimalis et al., 2014) is Saronikos Gulf, which includes the industrial zone of Elefsis Bay and Piraeus Port. Saronikos Gulf (mean depth ~100 m) is semi-enclosed marine area with poor water exchange, and is located on the western edge of the central Aegean Plateau, near the city of Athens (Fig. 1). Industrial and shipping activities are continuously degrading the environmental quality of the gulf. Domestic and industrial effluents were released untreated into the gulf until the early 1990s, when the Waste Water Treatment Plant (WWTP) of Athens was constructed on the islet of Psittalia. Currently, wastewater treatment reduces suspended solids and organic load by about 93% and total nitrogen by about 80% in comparison with influent loads (Simboura et al., 2014). A series of projects have been monitoring the macrozoobenthos of Saronikos Gulf during the last decades (e.g., Sheppard, 1977; Zarkanellas, 1979; Voutsinou-Taliadouri and Satsmadjis, 1982; Friligos and Zenetos, 1988; Nicolaidou et al., 1993; Simboura et al., 1995, 2005; Makra et al., 2001). Based on ongoing macroinvertebrate monitoring data, ecological quality classification tools such as the BENTIX index (Simboura and Zenetos, 2002) have ranked Elefsis Bay as "poor" quality environment, whereas the inner Saronikos ranked "moderate" status at stations close to Psittalia outfalls and "good" at stations more than 15 km from the outfalls (Simboura et al., 1995, 2005, 2014). Interestingly, a general environmental improvement has been detected by most of the benthic indices over the last twelve years that became especially evident after 2004, when the advanced secondary biological treatment plant of Psittalia WWTP was completed (Simboura et al., 2014).

We assessed Saronikos Gulf sediment quality by quantifying the status of benthic foraminiferal assemblages, which are significant components of living meiobenthic communities. Using the resulting data in the context of previous research in the gulf and elsewhere, a new index of ecosystem quality is presented that can be applied to meet the Marine Strategy Framework Directive – MSFD (2008/56/EC), in combination or complementary to macrofauna quality classification indices. The MSFD Directive is a more holistic directive than the WFD, requiring information on benthos, sediment, and the structural and functional benthic aspects (Cardoso et al., 2010; Borja et al., 2010; van Hoey et al., 2010). Therefore, a new index, the Foram Stress Index (FSI), is proposed for ecological and environmental assessment of the status of soft-bottom benthic meiofaunal communities.

2. Material and methods

2.1. Sample collection

Sediments were collected onboard the R/V Aegaeo from 12 sampling stations, two in Elefsis Bay and ten in Saronikos Gulf in February 2012, at water depths ranging from 20 to 94 m (Fig. 1; Table 1). The sampling stations network was based on the criterion of increasing distance from the Psittalia WWTP pipeline (point source of the effluents). The stations have been occupied regularly since 2000 and up to the present day to monitor the impact of the Psittalia WWTP. At each station, two replicate samples from different deployments were collected using a stainless steel box-corer of 1000 cm^2 sampling surface. Mean values from the two replicates are used in the present study.

2.2. Sediment analyses

Grain size, organic carbon (OC) and selected heavy metals were assessed in all samples. Grain-size analysis followed Folk's (1974) methodology. The sand fraction was separated by wet sieving using a 63 μm pore size sieve. The fine silt and clay fractions (mud) were analyzed in a Micromeritics Sedigraph 5100 X-ray grain-size analyzer. The organic carbon was determined using an elemental analyzer (Fisons EA-1108 CHN), following the procedures described by Verardo et al. (1990). The precision of the method was within 5%. Sediment heavy metal (Cr, Cu, Ni, Pb and Zn) and metalloid (As) contents (hereafter metals) were determined by X-Ray Fluorescence (XRF) using a Philips PW-2400 instrument, following the method described by Karageorgis et al. (2009). Metal contents were compared against the empirical sediment quality guidelines ERL (effects range – low) and ERM (effects range – median) (Long et al., 1995, 1998).

2.3. Benthic foraminiferal analyses

For benthic foraminiferal analysis, the sediment surface was sampled to a depth of 1 cm (mean sample size 10 cm^3) from the central, least disturbed part of each box core. The sample was immediately stained with rose Bengal dissolved in 70% ethanol (2 g l^{-1}) to facilitate distinguishing live from dead foraminiferal tests. In the laboratory, samples were washed over 125 μm sieve and dried. Subsamples of the >125 μm sediment fraction,

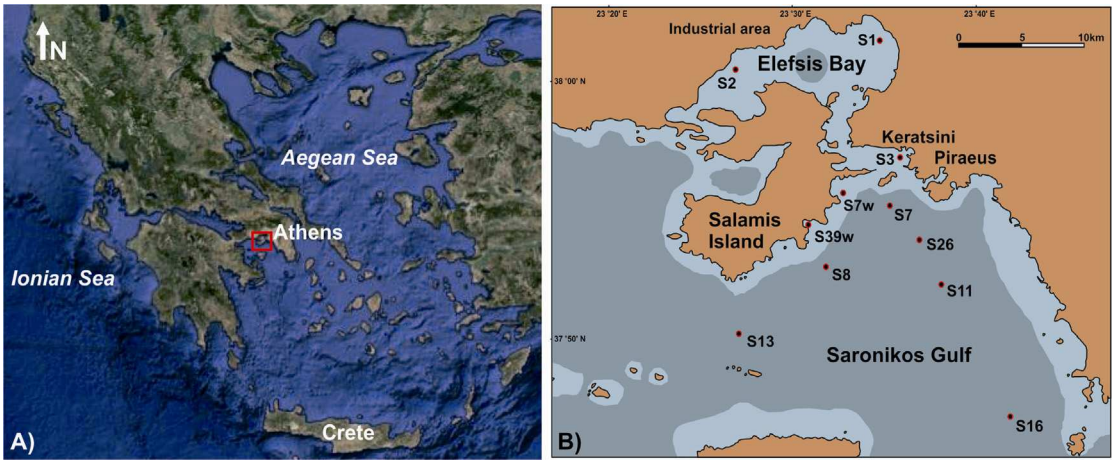


Fig. 1. (A) Map of the studied area, (B) locations of the sampled stations.

Table 1
Geographic coordinates, distance from the point source of the effluents and water depth from each sampled station.

Station	Latitude	Longitude	Distance from outfall (m)	Depth (m)	Mud%
S1	38° 01.83′	23° 34.83′		20	80
S2	38° 00.66′	23° 27.03′		33	96
S3	37° 57.30′	23° 36.00′	1845	29	29
S7	37° 55.42′	23° 35.45′	1790	65	91
S7w	37° 55.88′	23° 32.88′		30	25
S8	37° 53.00′	23° 32.00′	8040	90	94
S11	37° 52.36′	23° 38.30′	8488	74	34
S13	37° 50.41′	23° 27.30′	16,193	94	21
S16	37° 47.23′	23° 42.04′	19,430	90	19
S26	37° 54.10′	23° 37.08′	4880	70	52
S39w	37° 54.66′	23° 31.08′	4680	30	13

containing at least 300 living benthic foraminifers for each sample, were obtained using an Otto microsplitter. All subsamples were examined under a Leica APO S8 stereoscope and living (rose Bengal stained) specimens from each sample were picked and identified following the generic classification of Loeblich and Tappan (1987). From the resulting data set, several parameters were calculated including total and living foraminiferal absolute abundances (specimens cm^{-3}), and relative abundances of foraminiferal species, species richness (S), dominance (D) and Shannon–Wiener diversity (H) in the living assemblages.

Although total foraminiferal assemblages integrate information on the general conditions over a longer time period changes, including seasonal variations and productivity blooms (Scott and Medioli, 1980; Alve and Nagy, 1986; Buzas et al., 2002; Debenay et al., 2001, 2005), they may include allochthonous specimens and be biased by post-mortem processes (Alve and Murray, 1994). For a study such as this, in which the goal is to assess short-term environmental influences on soft-sediment assemblages, counts of living taxa are essential (Schönfeld et al., 2012).

2.4. Statistical analyses

Q-mode hierarchical cluster analysis (HCA) was used to estimate similarities in species composition among samples using SPSS (version 10.1) statistical software. Analysis was applied to 23 taxa that exceeded 5% of foraminiferal assemblage in at least one sample. Ward's method and Euclidean distance were used in the analysis. The data were logarithmically transformed to reduce the score and bias of more abundant species that may have otherwise masked the effect of less abundant species. Pearson's correlation coefficient (r) analysis was carried out to determine relationships between foraminiferal assemblages, and environmental parameters including water depth, percent mud, percent OC, and heavy metal and metalloid contents. All statistical analyses were performed using SPSS (version 10.1) statistical software.

2.5. Foram Stress Index (FSI)

A major goal in this study was to develop a foraminiferal index, the Foram Stress Index (FSI), to evaluate ecosystem quality for the Mediterranean Sea that could be used in the context of the Marine Strategy Framework Directive. The FSI is based on the Reef Assessment and Monitoring FORAM Index (FI), which was developed by Hallock et al. (2003) to indicate if ambient water quality can support calcifying animals that are dependent upon algal symbiosis (i.e., zooxanthellate corals and larger benthic foraminifers). The FI was initially developed using data from Caribbean and Western Atlantic reef sites (e.g., Hallock et al., 2003), and has subsequently been applied in other regions (Hallock, 2012), including the Australian Great Barrier Reef (Schueth and Frank, 2008; Uthicke and Nobes, 2008; Uthicke et al., 2010) and reefs off Brazil (Barbosa et al., 2009). Interestingly, FI has also been applied with success in subtropical estuaries (Carnahan et al., 2009; Narayan and Pandolfi, 2010), and even in the shallow rocky habitats of the eastern Mediterranean coastal environments (Koukousioura et al., 2011).

The context upon which the FI was developed (Hallock et al., 2003) included the ecological concepts of r and K selection (e.g., Pianka, 1970) and the model of foraminiferal response with distance from a point-source (Alve, 1995). Foraminifera are an exceptionally diverse group of protists, with benthic taxa occupying an equally diverse range of niches. There are taxa that can thrive in very clear, nutrient-poor waters because their algal symbionts can provide the energy that the foraminifer needs to find food and grow, often living for months or years and producing relatively large shells (e.g., Hallock, 1985). However, those foraminifers are easily outcompeted by smaller, faster-growing taxa if nutrient supplies provide abundant food for benthic organisms. At the other end of the spectrum, there are taxa that can tolerate environmental extremes including wide salinity fluctuations, low pH, hypoxia, and even concentrations of trace elements that are toxic to most eukaryotic organisms. In between there is a great diversity of benthic taxa that have adapted to the diverse range of niches that are found in marine environments. For the coastal zone, where human activities have most disrupted marine environments, the basic idea for an biotic index based on foraminifers is to translate that spectrum of possible niches to a linear scale ranging from physico-chemically relatively benign but food/nutrient-limited, to increasing availability of food and places to live, and then to environments with an over-abundance of food with consequent strong diurnal to seasonal fluctuations in pH and oxygen, and even intermittent hypoxia (see Alve, 1995; or Jorissen, 1999). In natural estuarine systems where organic matter can be in excess, there are often strong salinity fluctuations as well.

Our FSI, analogous to the FI, is mathematically simple and based upon recognizing that spectrum from food-limited environments with minimal physical or chemical stressors to very strong physical or chemical stressors that most taxa cannot tolerate, thus environments where there is plenty of food for taxa that are stress tolerant. Apparently, the major difference between FI, which was developed to include taxa that host symbiotic algae, and our developed FSI, is that the latter is not depth nor food/nutrient dependent therefore useful for ecological and environmental status assessment of soft-bottom benthic foraminiferal communities, including cases of naturally stressed environments such as lagoons or low salinity basins. We are utilizing primarily the heterotrophic foraminiferal taxa to characterize the gradient between extreme stress (abiotic) and unimpacted Mediterranean waters. Accordingly, we characterize two species groups, one "sensitive" (Sen) and one "stress tolerant" (Str), then scale the index based upon their relative abundances:

$$\mathrm{FSI} = (10 * \mathrm{Sen}) + (\mathrm{Str})$$

where Sen and Str represent the proportions of sensitive and stress-tolerant taxa, respectively. The (Sen) and (Str) species groups of FSI Index are based upon their ecological affinities, as follows:

(Sen): Species that are sensitive to organic enrichment are present in unpolluted ecosystems, disappearing or occurring in low proportions under organic enrichment. This group includes mainly epifaunal and sediment-surface-dwelling taxa, often associated with coarse-grained sediments. Included in this group are the few genera of symbiont-bearing foraminifers, including *Amphistegina*, *Peneroplis* and *Sorites*, that are widespread in the Mediterranean Sea, with preference for shallow marine waters with vegetation cover (e.g., Blanc-Vernet, 1969; Cimerman and Langer, 1991; Sgarrella and Moncharmont Zei, 1993; Murray, 2006;

Koukousioura et al., 2011). Other small epiphytic rotaliids, which are common in the Mediterranean infralittoral environments, such as *Rosalina*, *Cibicides*, *Asterigerinata*, *Cyclogyra*, *Discorbia*, *Elphidium*, *Reussela*, *Neoconorbina*, *Planorbulina* and *Lobatula*, are typical taxa in shallow-marine environments with well-oxygenated and vegetated or coarse-grained bottoms (Jorissen, 1987; Langer, 1988, 1993; Sgarrella and Moncharmont Zei, 1993; Frezza and Carboni, 2009; Barras et al., 2014). Finally, miliolids are rich in sandy, well oxygenated sediments, and appear to be highly sensitive to the increase of organic matter (Bizon and Bizon, 1984; Jorissen, 1988; Schmiedl et al., 2003; Barras et al., 2014; Li et al., 2014).

(Str): Stress-tolerant taxa may occur under unaltered conditions, but because they are tolerant of environmental variability, they can act opportunistically and increase their relative abundances compared with the more sensitive taxa under organic enrichment (Alve, 1995). This group includes mainly shallow and intermediate to deep infaunal species, associated with fine-grained sediments. The genus *Ammonia* is common in the shallow-marine sediments (e.g., Cimerman and Langer, 1991; Murray, 2006) and shows an opportunistic behavior and a high potential to survive under high organic matter and metal contents (Nikulina et al., 2008). Particularly, *Ammonia tepida* is commonly considered as a species tolerant of organic and chemical pollution (e.g., Yanko et al., 1999; Alve, 1995; Debenay et al., 2001; Armynot du Châtelet et al., 2004; Frontalini and Coccioni, 2008; Naeher et al., 2012). The genus *Bulimina* is a shallow to deep infaunal taxon (Jannink, 2001), tolerant of organic matter enrichment (e.g., Lutze and Colbourn, 1984; Jorissen et al., 1998; De Rijk et al., 2000; Fontanier et al., 2002), oxygen depletion (e.g., Lutze and Colbourn, 1984) and anthropogenic eutrophication (Mojtahid et al., 2006). Other opportunistic genera include *Bolivina*, *Melonis*, *Valvulineria*, *Cassidulina*, *Hyalinea*, *Hanzawaia*, *Globocassidulina*, *Rectuvigerina*, *Nonionella*, *Nonion* and *Fursenkoina*, which are common in foraminiferal assemblages associated with muddy bottoms with abundant organic matter and periodic oxygen deficiency (e.g., Jorissen, 1987; Barmawidjaja et al., 1995; Fontanier et al., 2002; Mojtahid et al., 2006, 2008; Barras et al., 2014).

Among agglutinated species, *Textularia agglutinans* is considered to be relatively tolerant of food-enriched and oxygen-deficient conditions (Barmawidjaja et al., 1995; Naeher et al., 2012). *Bigenerina nodosaria* is associated with deposition of fresh organic matter (Schmiedl et al., 2000; Goineau et al., 2011), conditions that often accompany pollution.

The FSI varies from 1 to 10, being 0 only when the sediment is azoic (all groups zero). The boundaries enabling the distinction of the five Ecological Quality Status (ECoQ) classes according to the WFD Directive are shown in Table 2.

The proposed values have been based on field observations (e.g., Armynot du Châtelet et al., 2004; Burone et al., 2006; Mojtahid et al., 2006; Bergamin et al., 2009; Cherchi et al., 2009; Hyams-Kaphzan et al., 2009; Denoyelle et al., 2010; Koukousioura et al., 2011; Barras et al., 2014), which include numerical data for (Str) frequencies in impacted environments (Table 3). Our data support an overall estimation of moderate ECoQ for percentages >50% concerning (Str) species. Therefore, according to the available literature data and our own observations, we consider the percentages contribution of stress tolerant species (Str) between 10 and 50%, for sites representing good ECoQ. Subsequently, the poor quality status corresponds to assemblages consisting of (Str) relative abundances over 90%, whereas the high quality status is characterized by (Str) less than 10%. To provide reference values in pristine environment (Table 3), we have applied FSI to soft bottom benthic foraminiferal assemblages of Andros island (Korthi gulf, FI > 4; Koukousioura et al., 2011); FSI values resulted as >9, confirming High/Good environmental status.

Table 2
Classification scheme of soft bottom foraminiferal habitats based on the Foram Stress Index.

Pollution classification	FSI	Boundary EQR value	Ecological Quality Status (ECoQ)
Normal/pristine	$9.0 \leq FSI \leq 10.0$	High/Good = 0.90	High
Slightly polluted, transitional	$5.5 \leq FSI < 9.0$	Good/Moderate = 0.55	Good
Moderately polluted	$2.0 \leq FSI < 5.5$	Moderate/Poor = 0.20	Moderate
Heavily polluted	$1.0 \leq FSI < 2.0$	Poor/Bad = 0.10	Poor
Azoic	Azoic conditions	–	Bad

3. Results

3.1. Sediment analyses

Grain-size analysis revealed the dominance of muddy sediments (silt and clay up to 50%) in the inner and western outlet of Elefsis Bay (stations S1 and S2) and in the central part of the sampling area (stations S7, S8 and S26).

Higher sediment organic carbon contents (2.2–3.1%) were observed in Elefsis Bay (stations S1 and S2) and close to Psittalia island (stations S7; 3.0%). The OC contents in the sediments from the other sampling stations were relatively low (<1.5%).

Arsenic contents varied from 7 to 29 $\mu g\,g^{-1}$, and Cr contents from 40 to 283 $\mu g\,g^{-1}$. For both As and Cr, the highest values were observed in Saronikos Gulf, near the WWTP, followed by Elefsis Bay (Table 4). As far as it concerns the other metals, nickel (17–125 $\mu g\,g^{-1}$), copper (10–140 $\mu g\,g^{-1}$), zinc (30–357 $\mu g\,g^{-1}$), and lead (11–138 $\mu g\,g^{-1}$), they exhibit their highest contents in Elefsis Bay (stations S1, S2; Table 4), followed by Saronikos Gulf, near the WWTP (stations S3, S7; Table 4). For all metals and As, the lowest contents were recorded in the southern sector of Saronikos Gulf (e.g., stations S11, S13, S16; Table 4), which communicates with the open sea.

Compared to the sediment quality guidelines (ERL and ERM), all metals show values higher than ERL at stations S1, S2, S3 and S7. However, out of these metals only Ni contents were greater than ERM threshold values (>51.6 $\mu g\,g^{-1}$) at stations S1, S2, S7, S8 and S26. According to Long et al. (1998), when ERMs are exceeded it is most likely that samples demonstrate toxicity. However, Karageorgis and Botsou (2014) have noted that the local metal (Cr and Ni) background values maybe naturally higher due to the weathering of metal-rich rock formations, thus contamination assessments should be conducted with caution and only toxicity tests could ultimately support incidence of sediment toxicity.

3.2. Benthic foraminiferal assemblages

The absolute abundances of total foraminiferal specimens were quite variable in the suite of samples, ranging from 101 (station S1) to 23,253 (station S7w) specimens cm^{-3}. Stained specimens (collected live) ranged from 0.12% (station S3) to 54% (station S2) of the total counts. Live-specimen absolute abundances fluctuated between 16 (station S3) and 954 (station S11) specimens cm^{-3}. Station S3 in Keratsini Bay was excluded from statistical analyses due to the low number of living specimens. A total of 55 living foraminiferal species and 40 genera were identified, including 39 hyaline, 13 porcellaneous and 3 agglutinated species (Appendix A, Plate 1).

Table 3
Summary of some published work dealing with evaluation of environmental conditions, mostly based on the foraminiferal assemblages of surface sediments.

Authors	Stress tolerant species (Str)	Interpreted environmental conditions	(Str) index species
Armynot du Châtelet et al., 2004	>50%	High polluted area	*Haynesina germanica*
Barras et al., 2014	0–46%	Faunas characterized by a high percentage of stress-tolerant species are indicative of fine-grained substrates often associated with high organic matter contents. Often, the predominance of stress-tolerant taxa is probably the result of natural conditions. However, in some cases it may be due to a super imposed anthropogenic impact	*Bulimina* spp., *Cancris auriculus*, *Nonion scaphum*,*Nonion depressulum*, *Nonionella turgida*, *Nonionella stella*, *Pseudoeponides falsobeccarii*, *Rectuvigerina phlegeri*, *Valvulineria bradyana*, *Leptohalysis scotti*
Bergamin et al., 2009	Increase of "pollution-tolerant species	High degree of environmental stress	*Ammonia tepida*, *Haynesina germanica*
Burone et al., 2006	>50%	Pure organic contamination	*Ammonia tepida*
Cherchi et al., 2009	>50%	Pollution increase	*Haynesina* spp.
Denoyelle et al., 2010	>60%	Strong environmental impact	*Bolivina striatula*, *Bulimina aculeata*, *Amphicoryna scalaris*, *Eggerella sp.*, *Bulimina marginata*, *Gyroidina* sp., *Bulimina costata*, *Uvigerina peregrina*, *Valvulineria bradyana*, *Bolivina dilatata*
Hyams-Kaphzan et al., 2009	30–70%	High dispersal of organic loads	*Ammonia tepida*
Koukousioura et al., 2011	Up to 20%	Influence of sewage pollution	*Ammonia* spp., *Elphidium* spp. (application of FORAM index in algal samples)
Present study (Andros island, Korthi gulf)	<10%	Pristine environment	*Ammonia* spp., *Elphidium* spp.
Mojtahid et al., 2006	About 80%	Zone of high stress/eutrophicated conditions	*Bulimina aculeata*, *Bulimina marginata*, *Textularia sagittula*, *Trifarina bradyi*, *Bolivina* spp.

Table 4
Heavy metal (Cr, Cu, Ni, Pb and Zn) and metalloid (As) contents ($\mu g\,g^{-1}$), ER-L and ER-M guideline values.

Station	As	Cr	Cu	Ni	Pb	Zn
S1	16	162	116	86	138	355
S2	15	170	140	125	97	357
S3	24	206	101	45	111	246
S7	29	283	127	100	126	280
S7w	11	126	24	34	26	57
S8	8	142	26	83	39	74
S11	11	57	12	26	21	36
S13	10	60	12	32	21	33
S16	8	40	71	17	23	305
S26	15	120	25	54	42	88
S39w	7	63	10	19	11	30
ER-L	8.2	81	34	20.9	46.7	150
ER-M	70	370	270	51.6	218	410

Species richness varied from $S = 8$ (station S2) to $S = 28$ (station S13), with an average of 17 species per sample. Dominance ranged from $D = 0.06$ (stations S13) to $D = 0.31$ (station S2); higher values were observed particularly in Elefsis Bay. In general, the Dominance index presents an opposite trend in respect to the Shannon–Wiener index ($H_{max} = 3.12$ at station S13 and $H_{min} = 1.72$ at station S1). The highest FSI value (FSI = 0.87) was observed at station S7w and the lowest (FSI = 0.12) at station S2.

Fourteen foraminiferal taxa made up more than 70% of the total assemblage: *Bulimina* spp., *Bolivina* spp., *Nonion fabum*, *Ammonia* spp., *Peneroplis* spp., *Elphidium* spp., *Rosalina* spp., *Planorbulina mediterranensis*, *Asterigerinata mammila*, *Lobatula lobatula*, *Melonis pompilioides*, *Cassidulina carinata*, *Rectuvigerina phlegeri* and miliolids. All foraminiferal species encountered are listed alphabetically in Appendix A.

Bulimina was the most abundant genus, comprising about 18% on average of the living foraminiferal abundance and was represented by *B. elongata* (max = 51% in station S2), *B. aculeata* (max = 31% in station S2) and *B. marginata* (max = 8% in station S2). *Bolivina* ssp. (average abundance = 8%), *N. fabum* (average abundance = 7%), *M. pompilioides* (average abundance = 2%) *C. carinata* (average abundance = 2%) and *R. phlegeri* (average abundance = 2%) exhibited more or less similar distribution patterns with maximum abundance in the central part of the sampling area (stations S7, S8, S13 and S26). *Ammonia*, represented mainly by *A. tepida*, was a significant element with an average abundance up to 18%, in Elefsis Bay (stations S1 and S2). *Rosalina* spp. (*R. bradyi* and *R. macropora*) were not found in Elefsis Bay, whereas they occurred commonly (8–11%) samples from the eastern part of Salamis coasts (stations S7w and S39w). *P. mediterranensis* showed a similar trend, reaching 16% at station S7w. Other important epiphytic taxa, including *Elphidium* spp. (mainly *E. advenum*), *Asterigerinata mammilla* and *L. lobatula*, somewhat increased toward the eastern sector of the sampling area (stations S11, S13 and S16). Among porcellaneous taxa, *Peneroplis* spp. (mainly *P. pertusus*), were found only in the eastern part of Salamis coasts (stations S7w and S39w), being the dominant taxon (>20%) of the living fauna. Miliolids were represented mostly by *Triloculina*, *Quinqueloculina* and *Adelosina* and comprised up to 29, with higher relative abundance values (more than 20%) in the eastern part of Salamis coasts (stations S7w and S39w) and toward the open sea area (stations S11 and S16).

3.3. Statistical analyses

Q-mode hierarchical cluster analysis (Fig. 2) was applied to determine the degree of similarity in foraminiferal species composition among the studied stations, intending to explore possible relationships between foraminiferal distributional patterns and geographical location, substrate, OC and metal contents. The resulting dendrogram distinguished three clusters: Cluster 1 included the two stations in Elefsis Bay (S1 and S2). Cluster 2 grouped four somewhat offshore stations (S7, S8, S13 and S26), whereas Cluster 3 is represented by two Salamis Island stations (S7w and S39w) and two open gulf stations (S11 and S16).

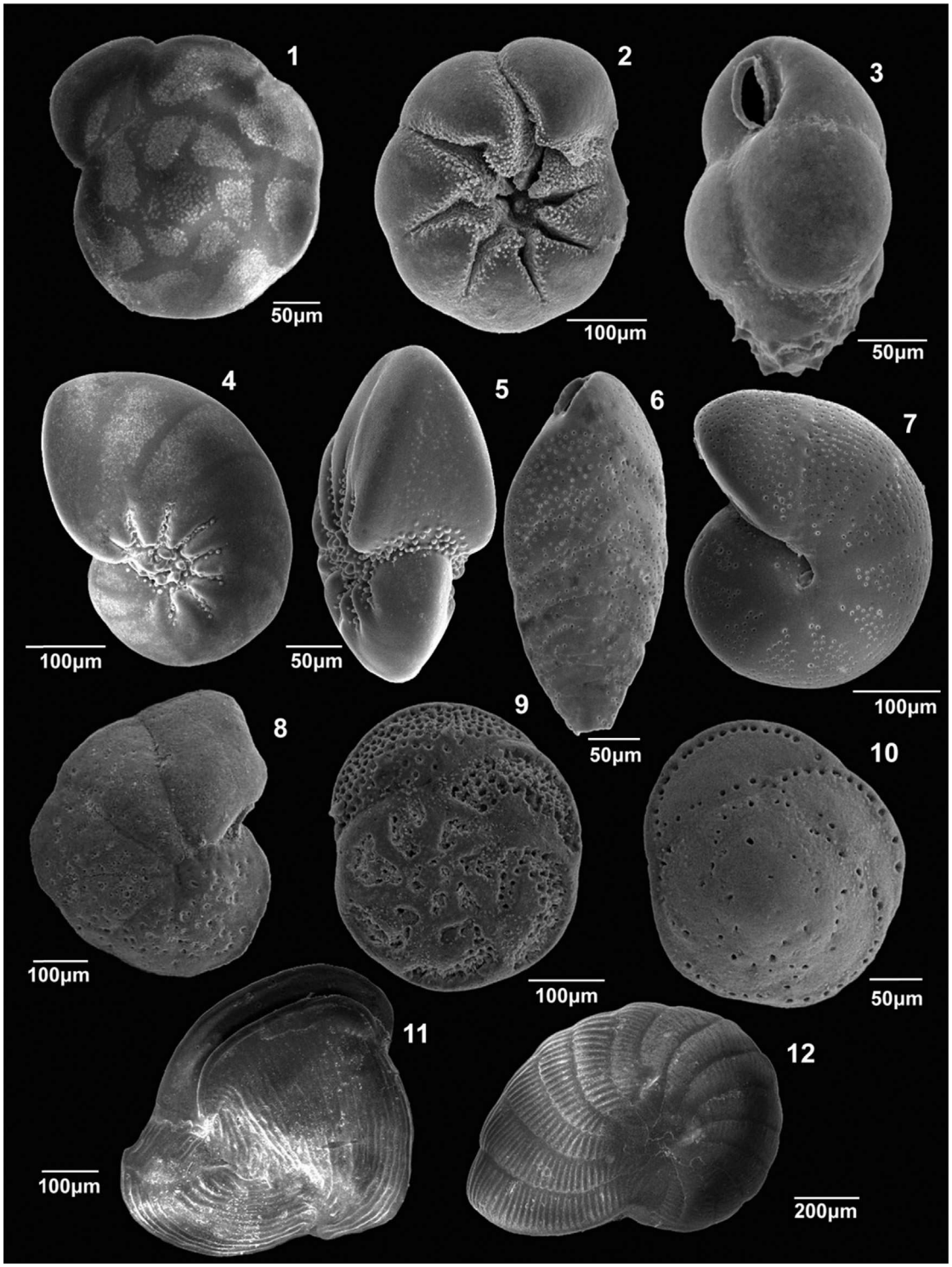


Plate 1. Foraminiferal specimens. 1 – *Ammonia tepida* (Cushman), spiral side, S1; 2 – *Ammonia tepida* (Cushman), umbilical side, S2; 3 – *Bulimina aculeata* d'Orbigny, lateral view, S2; 4 – *Nonion fabum* (Fichtel & Moll), side view, S7; 5 – *Nonion fabum* (Fichtel & Moll), peripheral view, S8; 6 – *Bolivina sphatulata* (Williamson), lateral view, S7; 7 – *Melonis pompilioides* (Fichtel & Moll), side view, S26; 8 – *Lobatula lobatula* (Walker & Jacob), umbilical side, S11; 9 – *Rosalina bradyi* (Cushman), spiral side, S39w; 10 – *Asterigerinata mamilla* (Williamson), spiral side, S16; 11 – *Vertebralina striata* d'Orbigny, side view, S39w; 12 – *Peneroplis pertusus* (Forskål), side view, S7w.

Foraminiferal assemblages were compared to abiotic parameters using the Pearson's correlation coefficient *r* (Table 5). Foraminiferal absolute abundance revealed no significant correlation with the abiotic parameters. Species richness, Shannon–Wiener index and FSI negatively correlated with percent mud, organic carbon and with most of the metals, while Dominance showed positive correlation with the same parameters. *A. tepida*, *Bolivina spathulata*, *M. pompilioides* and *Bulimina* spp. positively correlated with several heavy metal (Cu, Ni, Pb and Zn) and metalloid (As) contents, whereas *L. lobatula*, *Asterigerinita mamilla*, *Rosalina* spp. and miliolids displayed significant negative values. Moreover, *Bulimina* spp. exhibited a direct relationship with organic carbon, whereas *A. mamilla*, *Rosalina* spp. and miliolids negatively correlated with percent mud.

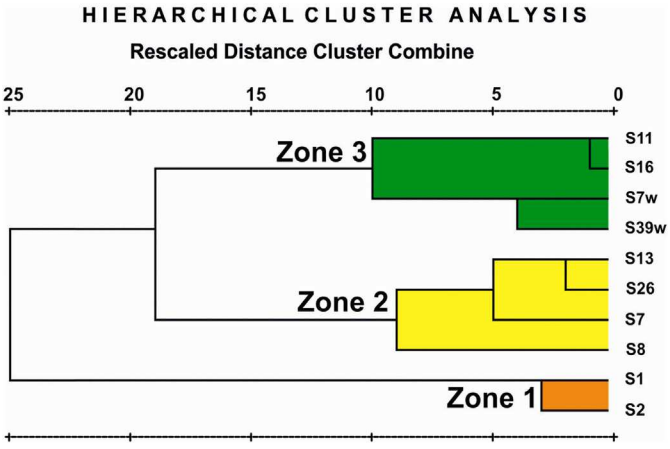


Fig. 2. Dendrogram resulting from a Q-mode hierarchical cluster analysis (Ward's method, Euclidean distances as a similarity index) of studied foraminiferal stations.

3.4. Comparison of FSI and BENTIX in Saronikos Gulf

To evaluate the effectiveness of the proposed FSI, we have compared our results with the BENTIX index, which is a biotic benthic index used to assess the ecological and environmental status of soft-bottom macrobenthic communities under the requirements of EU policies. Specifically for the needs of the MSFD, the BENTIX index has been proposed to be used in combination with diversity indices (Simboura et al., 2012, 2014). For the comparison of both FSI and BENTIX indices, a scatter graph of the normalized Ecological Quality Ratios was constructed (Fig. 3). The EQRs normalization and the agreement level calculation among the indices followed the methodology applied during the Mediterranean Geographical Intercalibration Group (MED-GIG) exercise (van de Bund et al., 2008) under the WFD.

4. Discussion and conclusions

According to cluster analysis (Fig. 2), the stations of Saronikos Gulf can be divided into three zones, each of which is characterized by a distinct foraminiferal assemblage related to certain values of the abiotic parameters (Table 6).

Zone 1 includes the two samples from shallow, semi-enclosed Elefsis Bay. Sediments are very muddy (80–96%), rich in OC (2.3–3.1%), and have metal contents that exceed established Effects-Range minima. The sparse foraminiferal assemblage is dominated by the stress tolerant *A. tepida* and *Bulimina* spp. (mostly *B. elongata*), followed by *B. aculeata* and *B. marginata*. Dominance values are relative high (D = 0.24 and 0.31), whereas species richness (S = 8 and 9) and Shannon–Wiener indices (H = 1.42 and 1.72) are relatively low. FSI mean value is 1.60 ± 0.25 with FSI EQR values ranging between 0.12 and 0.20 (Table 7); therefore, Zone 1 is classified as having a "poor" ecological status.

Zone 2 stations (S7, S8, S26, S13) were collected in the central and the western sector of the sampling area. Sediments are characterized by high variability, with mud percentages (21–94%), OC levels (0.7–3.0%) and metal contents spanning the range between Zone 1 and Zone 3 values. *Bolivina* spp. and *N. fabum* dominate this Zone. Several other opportunistic species, such as *C. carinata*, *R. phlegeri* and *B. elongata*, are found with epiphytic *Rosalina* spp. and *Elphidium* spp., though the absolute abundance of foraminiferal specimens remained quite low (<100/cm^{-3}) The addition of the more sensitive epiphytic taxa to the assemblage raised the Shannon–Wiener index (H = 2.31–3.02) and reduced dominance (D = 0.06–0.15), compared with Zone 1. FSI mean value is 3.59 ± 0.31 with FSI EQR values ranging between 0.24 and 0.53 (Table 7), indicating moderately polluted environmental status.

Zone 3 included two samples from near the coast of Salamis stations (S7w, S39w) and two from the eastern sector of the sampling area toward the open sea (stations S11, S16). Sediments are characterized by lower mud percentages (13–34%), OC levels (<1.5%) and metal contents, compared to Zones 1 and 2. The absolute abundance of foraminiferal specimens is roughly 6–8 times higher than in the other zones, and includes miliolid species, the algal symbiont *P. pertusus* and a number of small epiphytics, such as *A. mammila*, *P. mediterranensis*, *Rosalina* spp., and *Elphidium* spp. While overall diversity (H = 2.35–3.02) and dominance (D = 0.06–0.11) are very similar to those for Zone 2 assemblages, the FSI mean value is 8.10 ± 0.16 with FSI EQR values ranging between 7.27 and 8.67 (Table 7) indicates good environmental status for Zone 3.

Table 5
Correlation matrix (Pearson) among foraminiferal assemblage and abiotic parameters. r values greater than |0.63| are significant with 95% probability.

	Depth	Mud	OC	As	Cr	Cu	Ni	Pb	Zn
Species richness *S*	0.61	**−0.72**	**−0.81**	−0.50	**−0.71**	**−0.86**	**−0.79**	**−0.82**	**−0.81**
Dominance *D*	−0.61	**0.68**	**0.72**	0.20	0.44	**0.73**	**0.79**	**0.67**	**0.68**
Shannon–Wiener index *H*	**0.64**	**−0.74**	**−0.78**	−0.32	−0.58	**−0.78**	**−0.82**	**−0.74**	**−0.73**
FSI EQR	0.04	**−0.91**	**−0.65**	−0.50	**−0.65**	−0.63	**−0.90**	**−0.75**	−0.52
A. tepida	−0.57	0.38	0.41	0.16	0.22	0.51	0.39	**0.67**	0.56
A. mamilla	0.37	**−0.63**	−0.52	−0.44	**−0.67**	−0.25	**−0.68**	−0.52	−0.04
B. spathulata	0.25	0.22	0.25	**0.77**	0.50	0.31	0.20	0.45	0.25
B. striatula	0.49	0.49	−0.07	−0.04	0.26	−0.15	0.30	−0.02	−0.21
B. aculeata	−0.45	0.54	**0.67**	0.14	0.29	**0.67**	**0.70**	0.49	0.61
B. elongata	−0.52	**0.69**	**0.85**	0.43	0.55	**0.85**	**0.84**	**0.79**	**0.76**
B. marginata	−0.55	0.40	**0.70**	0.54	0.58	0.61	0.56	0.58	0.53
C. carinata	0.38	−0.09	−0.38	−0.03	−0.18	−0.37	−0.16	−0.25	−0.32
Elphidium spp.	0.53	0.60	−0.41	−0.49	−0.12	−0.42	−0.22	−0.45	−0.26
L. lobatula	0.62	−0.55	**−0.64**	−0.42	**−0.68**	**−0.67**	**−0.64**	**−0.65**	−0.60
M. pompilioides	0.37	0.12	0.15	**0.65**	0.42	0.08	0.15	0.22	−0.06
N. fabum	**0.66**	0.23	−0.07	0.38	0.33	−0.09	0.15	0.03	−0.17
Peneroplis spp.	−0.54	−0.52	−0.24	−0.33	−0.21	−0.40	−0.43	−0.40	−0.44
P. mediterranensis	−0.54	−0.62	−0.33	−0.39	−0.32	−0.42	−0.56	−0.43	−0.40
R. phlegeri	0.42	0.48	−0.11	−0.15	0.18	−0.19	0.27	−0.07	−0.22
Rosalina spp.	0.21	**−0.65**	−0.62	−0.35	−0.35	**−0.76**	**−0.70**	**−0.75**	**−0.78**
Miliolids	0.22	**−0.86**	−0.61	−0.50	**−0.69**	−0.57	**−0.86**	**−0.74**	−0.48
Agglutinants	**0.64**	−0.30	−0.26	−0.04	−0.27	−0.08	−0.40	−0.27	0.03

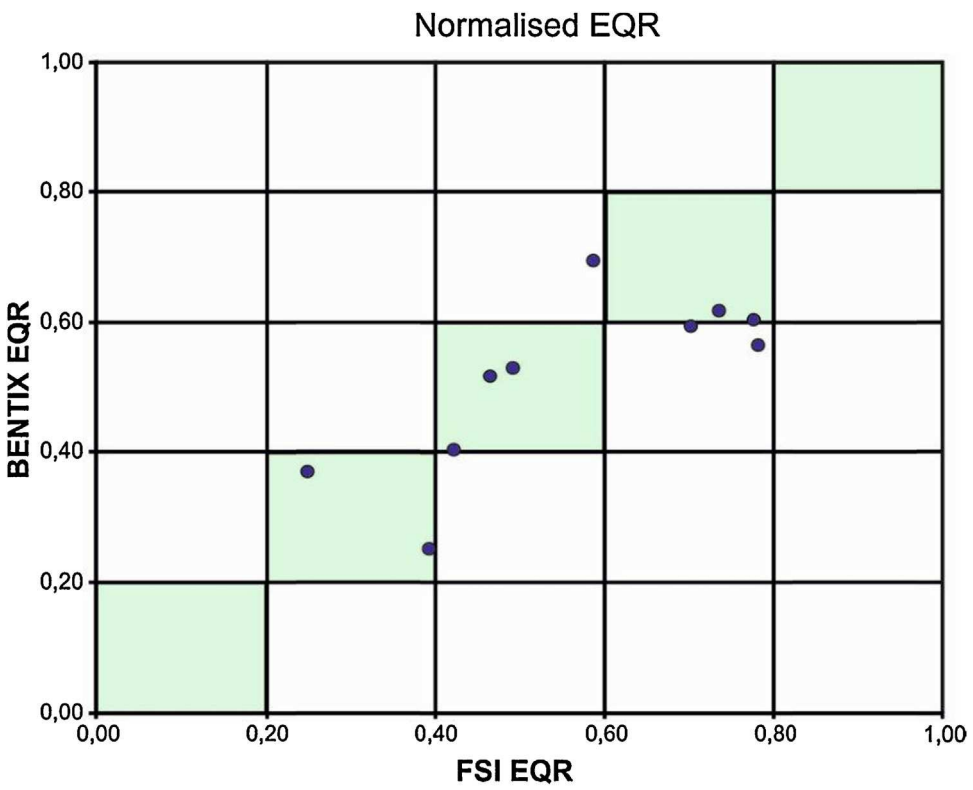


Fig. 3. Plot of the normalized EQR correlations (scatter graph) among FSI and BENTIX indices in Saronikos Gulf.

Table 6
Average abundance of common benthic species (numbers in bold indicate relative abundance >5%), minimum and maximum values of foraminiferal and abiotic parameters in each Zone. Mean and standard deviation of heavy metals and metalloids contents, along with ERL and ERM sediment quality guideline thresholds.

	Zone 1	Zone 2	Zone 3	ERL	ERM
A. tepida	**23.3**	0.3	0.8		
A. mamilla	0.0	0.9	10.3		
B. spathulata	2.7	**7.2**	1.4		
B. striatula	0.0	11.2	0.0		
B. aculeata	**19.8**	0.3	0.0		
B. elongata	**36.8**	**5.8**	0.4		
B. marginata	**6.7**	3.1	2.3		
C. carinata	0.0	**5.4**	1.3		
Elphidium spp.	2.7	**7.0**	**5.6**		
L. lobatula	0.0	3.4	**4.3**		
M. pompilioides	0.0	4.9	0.0		
N. fabum	0.5	**15.5**	3.6		
Peneroplis spp.	0.0	0.0	**12.3**		
P. mediterranensis	1.8	0.8	**9.7**		
R. phlegeri	0.0	**5.6**	0.4		
Rosalina spp.	0.0	**5.8**	**7.9**		
Miliolids	1.4	**8.9**	**26.0**		
Agglutinants	0	2.5	4.9		
Absolute abundance (specimens cm^{-3})	36–102	33–86	204–954		
Species richness *S*	8–9	13–28	17–27		
Dominance *D*	0.24–0.31	0.06–0.15	0.06–0.11		
Shannon–Wiener index *H*	1.4–1.7	2.3–3.1	2.5–3.0		
FSI	1.2–1.9	2.4–5.3	7.3–8.7		
Depth (m)	20–33	65–94	30–90		
%Mud	80–95	21–94	13–34		
%OC	2.3–3.2	0.73–3.0	0.57–1.5		
As ($\mu g\,g^{-1}$)	15–16	8–29	7–11		
Mean	15	15	9	8.2	70
Std. deviation	0.71	9.5	2.1		
Cr ($\mu g\,g^{-1}$)	162–170	60–283	40–126		
Mean	166	151	71	81	370
Std. deviation	5.7	94	38		
Cu ($\mu g\,g^{-1}$)	116–140	12–127	10–71		
Mean	128	47	29	34	270
Std. deviation	17	53	29		
Ni ($\mu g\,g^{-1}$)	86–125	32–100	17–34		
Mean	105	67	24	20.9	51.6
Std. deviation	28	30	7.7		
Pb ($\mu g\,g^{-1}$)	97–138	21–126	11–26		
Mean	117	57	20	46.7	218
Std. deviation	29	47	6.5		
Zn ($\mu g\,g^{-1}$)	355–357	33–280	30–305		
Mean	356	119	107	150	410
Std. deviation	1.41	110	133		

Table 7
Comparison between the FSI and the BENTIX index in Saronikos Gulf.

Stations	FSI	FSI EQR	ECoQ-FSI	BENTIX	BENTIX EQR	ECoQ-BENTIX
S1	1.96	0.20	P	2.12	0.35	P
S2	1.24	0.12	P	2.44	0.41	P
S7	3.13	0.31	M	3.08	0.51	M
S7w	8.67	0.64	G	3.31	0.55	M
S8	2.37	0.23	M	2.54	0.42	M
S11	7.27	0.52	G	3.45	0.58	M
S13	5.26	0.32	M	3.96	0.66	G
S16	7.86	0.58	G	3.57	0.60	G
S26	3.59	0.36	M	3.14	0.52	M
S39w	8.58	0.66	G	3.50	0.58	G

In a previous study, depth and sediment composition accounted significantly for the variance of diversity and species richness across a large soft-bottom benthic communities metadata set from the Hellenic marine environment and were used as factors for the definition of specific ecotypes and indices thresholds within them (Simboura et al., 2012). Yet, when comparing all calculated foraminiferal parameters, diversity indices such as the species richness, dominance and Shannon–Wiener can detect severe stress or pollution, but are much less indicative of the functional structure of

foraminiferal assemblage. A major limitation of these indices is that they do not consider the role of the species in the community (e.g., Debenay et al., 2005). A variety of studies have shown that diversity indices do not show a monotonic response to stress gradients. For example, in the Mediterranean Sea where unimpacted waters tend to be very clear and nutrient-poor (e.g., Koukousioura et al., 2011), diversity indices can be highest in cases of moderate disturbance and dominance can increase in the cleanest waters (Subida et al., 2012; Simboura and Argyrou, 2010). Thus, biotic indices that reflect ecological function and not simply mathematical function, can better reflect the ecological status of benthic communities. Our results indicate that, although Zones 2 and 3 have similar diversities, there are obvious differences in their species composition (Fig. 4). There is also a substantial difference in absolute abundances. Thus, the FSI that is based on an ecological understanding of foraminiferal taxa, provides a sensitive indicator of environmental and stress gradients. The FSI uncertainties in our Saronikos Gulf case study range between ~0.15 and 0.30 (Table 7); values may be considered useful for the future monitoring actions of the certain and/or similar areas.

Interestingly, the comparison between the FSI and BENTIX indices presents a discrepancy only in the Good/Moderate area, with S7w and S11 stations being classified as in Good ECoQ using FSI; a Moderate ECoQ has been assessed when applying the BENTIX index. An opposite result was produced for station S16 (Table 7).

Previous studies have shown that that greatest variability can be in the areas of intermediate impact (e.g., Hallock et al., 2003; Carnahan et al., 2009; Koukousioura et al., 2011), because natural variability in local environmental conditions can reduce or amplify stressors. Moreover, because macrobenthos will respond to environmental variability at different rates than meiobenthos, the most discrepancies can be anticipated in areas of intermediate impact. The sites where there is discrepancy between the indices are sites with relatively low percent mud, indicating at least intermittent exposure to currents. For example, intermittently strong currents where there is relatively abundant OC could affect macro- and meio-benthos quite differently.

Predictably, the FSI and BENTIX indices both classified the Elefsis Bay stations S1 and S2 as poor ECoQ. Stations S7, S8, S26, in the central sector of the sampling area, have been classified as moderately polluted by both indices, whereas the stations S16, S39w display Good ECoQ. Thus, a high degree of agreement (70%) is observed between the FSI and BENTIX indices (Fig. 3). As the two indices are well validated by correlations with sediment abiotic factors, their combined use is suggested as a specific tool for both benthic ecosystem and sediment-quality status assessments.

Acknowledgments

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Appendix A. Living benthic foraminiferal species identified in this study

Adelosina cliarensis (Heron-Allen & Earland, 1930)
Adelosina longirostra (d'Orbigny, 1826)
Ammonia beccarii (Linné, 1758)
Ammonia tepida (Cushman, 1926)
Amphistegina lobifera Larsen, 1976
Asterigerinata mamilla (Williamson, 1858)
Bigenerina nodosaria d'Orbigny, 1826
Bolivina sphatulata (Williamson, 1858)
Bolivina striatula Cushman, 1922
Buccella frigida (Cushman, 1922)
Bulimina aculeata d'Orbigny 1826
Bulimina elongata d'Orbigny, 1846
Bulimina marginata d'Orbigny, 1826
Cassidulina carinata Silvestri, 1896
Cibicides refulgens de Montfort, 1808
Cyclogyra planorbis (Schultze, 1853)
Discorbia valvulinerioides Sellier de Civrieux, 1977
Elphidium advenum (Cushman, 1922)
Elphidium complanatum (d'Orbigny, 1839)
Elphidium crispum (Linné, 1758)
Elphidium fichtelianum d'Orbigny, 1846
Elphidium granosum (d'Orbigny, 1826)
Elphidium macellum (Fichtel & Moll, 1798)
Globocassidulina subglobosa (Brady, 1881)
Hanzawaia boueana (d'Orbigny, 1846)
Hyalinea balthica (Schröter, 1783)
Lagena spp.
Lenticulina spp.
Lobatula lobatula (Walker & Jacob, 1798)
Melonis pompilioides (Fichtel & Moll, 1798)

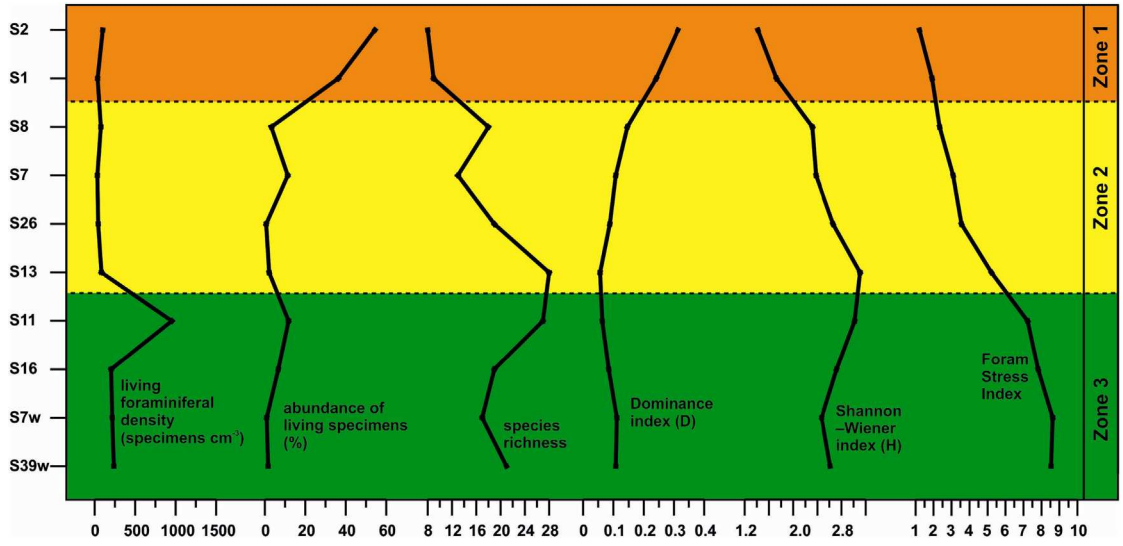


Fig. 4. FSI against absolute and relative abundance of living Foraminifera, species richness, Dominance and Shannon–Wiener indices in Saronikos Gulf.

Miliolinella subrotunda (Montagu, 1803)
Neoconorbina terquemi (Rzehak, 1888)
Nodosaria spp.
Nonionella turgida (Williamson, 1858)
Nonion fabum (Fichtel & Moll, 1798)
Peneroplis pertusus (Forskål, 1775)
Peneroplis planatus (Fichtel & Moll, 1798)
Planorbulina mediterranensis d' Orbigny, 1826
Quinqueloculina berthelotiana d'Orbigny, 1839
Quinqueloculina laevigata d'Orbigny, 1826
Quinqueloculina seminula (Linné) Cushman, 1930
Rectuvigerina phlegeri Le Calvez, 1959
Reussella spinulosa (Reuss, 1850)
Robertina spp.
Rosalina bradyi (Cushman, 1915)
Rosalina macropora (Hofker, 1951)
Sigmoilina costata Schlumberger, 1893
Sorites orbiculus (Forskål, 1775)
Spirotextularia spp.
Stainforthia spp.
Textularia agglutinans d'Orbigny, 1839
Triloculina tricarinata d'Orbigny, 1826
Triloculina trigonula (Lamarck, 1804)
Valvulineria bradyana (Fornasini, 1900)
Vertebralina striata d'Orbigny, 1826

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